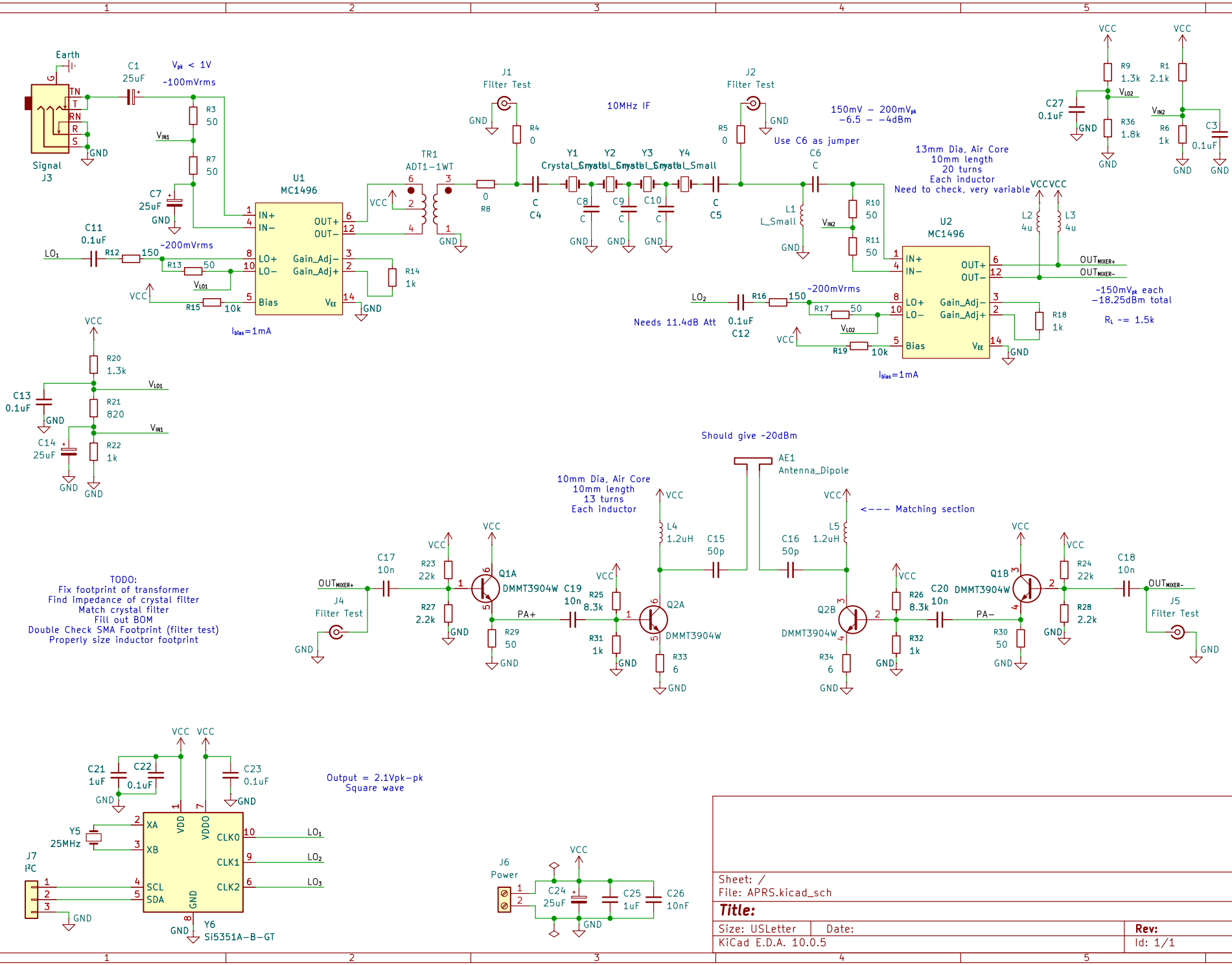

Earth
Signal
J3
C1
25uF
V_{pk} < 1V
~100mVrms
R3
50
R7
50
V_{IN1}
C7
25uF
GND
C11
0.1uF
LO_1
R12
150
~200mVrms
R13
50
V_{LO1}
R15
10k
U1
MC1496
IN+
IN−
LO+
LO−
Bias
OUT+
OUT−
Gain_Adj−
Gain_Adj+
V_{EE}
VCC
R14
1k
I_{bias}=1mA
TR1
ADT1−1WT
R8
0
J1
Filter Test
R4
0
C4
10MHz IF
Y1
Crystal_Small
Y2
Crystal_Small
Y3
Crystal_Small
Y4
Crystal_Small
C8
C9
C10
C5
J2
Filter Test
R5
0
Use C6 as jumper
C6
150mV − 200mVpk
−6.5 − −4dBm
13mm Dia, Air Core
10mm length
20 turns
Each inductor
Need to check, very variable
L1
L_Small
R10
50
R11
50
V_{IN2}
U2
MC1496
L2
4u
L3
4u
OUT_{MIXER+}
$OUT_{MIXER−}$
~150mVpk each
−18.25dBm total
R_L ~= 1.5k
LO_2
R16
150
~200mVrms
R17
50
V_{LO2}
R19
10k
Needs 11.4dB Att
0.1uF
C12
R18
1k
I_{bias}=1mA
R9
1.3k
R1
2.1k
C27
0.1uF
R36
1.8k
R6
1k
C3
0.1uF
R20
1.3k
R21
820
R22
1k
C13
0.1uF
C14
25uF
Should give ~20dBm
AE1
Antenna_Dipole
10mm Dia, Air Core
10mm length
13 turns
Each inductor
L4
1.2uH
C15
50p
L5
1.2uH
C16
50p
<--- Matching section
C17
10n
R23
22k
R27
2.2k
Q1A
DMMT3904W
C19
10n
R25
8.3k
R31
1k
Q2A
DMMT3904W
R29
50
R33
6
PA+
Q2B
DMMT3904W
R26
8.3k
R32
1k
R34
6
C20
10n
PA−
Q1B
DMMT3904W
R24
22k
R28
2.2k
R30
50
C18
10n
OUT_{MIXER+}
$OUT_{MIXER−}$
J4
Filter Test
J5
Filter Test
TODO:
Fix footprint of transformer
Find impedance of crystal filter
Match crystal filter
Fill out BOM
Double Check SMA Footprint (filter test)
Properly size inductor footprint
C21
1uF
C22
0.1uF
C23
0.1uF
Y5
25MHz
J7
I²C
XA
XB
SCL
SDA
VDD
VDDO
GND
CLK0
CLK1
CLK2
LO_1
LO_2
LO_3
Y6
Si5351A−B−GT
Output = 2.1Vpk−pk
Square wave
J6
Power
C24
25uF
C25
1uF
C26
10nF
Sheet: /
File: APRS.kicad_sch
Title:
Size: USLetter
Date:
Rev:
KiCad E.D.A. 10.0.5
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